Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms
B. 2.8 amps, 3.2 ohms
C. 3.75 amps, 3.2 ohms
D. 4 amps, 3 ohms
E. 3.33 amps, 3.6 ohms
F. 5 amps, 2.4 ohms
G. 4.5 amps, 2.67 ohms
H. 2.2 amps, 5.45 ohms
I. 2.5 amps, 4.8 ohms
J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>
To determine the required current and the corresponding resistance, use OhmâĢĻs law for the load resistor.
- Power dissipated by the bulb: P = V I = 40 W.
- Voltage across its terminals: V = 12 V.
- Current: I = P/V = 40 / 12 âīĪ 3.33 A.
- Resistive power (P_r) of the bulb: P_r = V ^ 2 / R = 12 ^ 2 / R = 40 W.
- Calculate: R = 144 / 40 = 3.6 Î©.

Thus, the bulb requires 3.33 A at 3.6 Î©.

Correct answer: E. 3.33 amps, 3.6 ohms.
</think>
<answer>
E. 3.33 amps, 3.6 ohms
</answer>